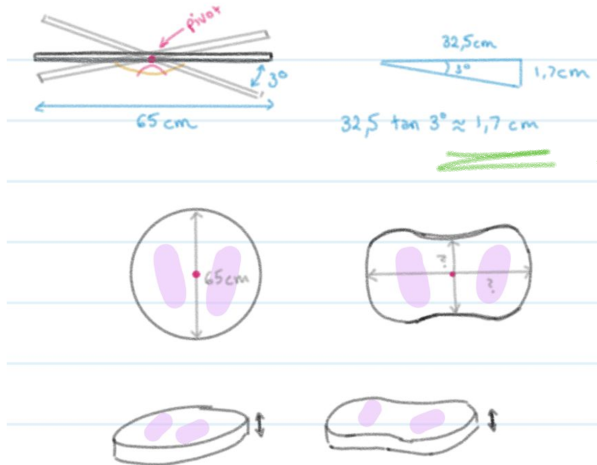


Design

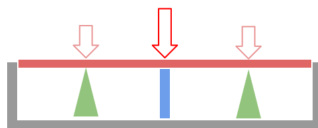


→ very little mechanically needed so height will be dependent on internal components

↳ lower is lower risk of buckling + safer

INNOVATION and BUSINESS comments? any research on this?

Force Analysis



Load bearing components

- Platform
- Pivot point(s)
- Actuators

So, for these → bending, shear, compression and buckling calculations

Maximum user mass : 100 - 120 kg ?

Design FOS : ~3

Design load → ~ 3.5 kN

static vs dynamic load (stepping, no jumping :))

Materials

Platform : wood, engineering plastics, steel [durable (+pretty)]

Pivot point(s) : engineering plastics, steel

Actuators : ?

Other parts (frame and non- load bearing components [sensor mounts etc.]) : 3D printed, plywood [light+cheap (+pretty)]

MECHANICAL ☺

and what about \$\$\$

MECHATRONICS :)